

Bachelor of Science with Honors in Mathematical Physics

Bilal Irfan Ansari, Class of 2031

Degree Overview

Year 1	
Fall Term	**Winter Term**
MATH 117 - Honors Calculus 1	MATH 118 - Honors Calculus 2
MATH 127 - Honors Linear Algebra 1	MATH 227 - Honors Linear Algebra 2
PHYS 144 - Newtonian Mechanics	PHYS 181 - Relativity, Electricity and Magnetism
CMPUT 174 - Introduction to Computation 1	CMPUT 175 - Introduction to Computation 2
STAT 151 - Introduction to Applied Statistics 1	CMPUT 267 - Machine Learning 1
-	CMPUT 272 - Formal Systems and Logic in CS
Spring Term	**Summer Term**
ENGL 102 - Introduction to Critical Analysis	ENGL 103 - Case Studies in Research
ECE 209A - Fundamentals of Electrical Engineering	ECE 209B - Fundamentals of Electrical Engineering
Year 2	
Fall Term	**Winter Term**
MATH 217 - Honors Advanced Calculus 1	MATH 317 - Honors Advanced Calculus 2
MATH 327 - Algebra 1	MATH 329 - Algebra 2
MA PH 251 - Differential Equations for Physics	MATH 432 - Intermediate Differential Equations
PHYS 244 - Classical Mechanics 1	PHYS 234 - Introductory Computational Physics
PHYS 295 - Experimental Physics 1	PHYS 271 - Introduction to Modern Physics
BIOPH 201 - Introduction to Biophysics	CMPUT 204 - Algorithms 1
PHYS 399 - Research Experience in Physics	PHYS 399 - Research Experience in Physics
Spring Term	**Summer Term**
WRS 101 - Exploring Writing	NS 201 - Indigenous Canada
ECON 101 - Introduction to Microeconomics	ECON 102 - Introduction to Macroeconomics
Year 3	
Fall Term	**Winter Term**
MATH 411 - Honors Complex Variables	MATH 417 - Honors Real Variables 1
MATH 448 - Introduction to Differential Geometry	MATH 447 - Elementary Topology
MA PH 343 - Classical Mechanics 2	MA PH 364 - Group Theory in Physics
PHYS 310 - Thermodynamics and Kinetic Theory	MA PH 451 - Mathematical Methods for Physics 2
PHYS 362 - Optical Physics	PHYS 311 - Statistical Physics
PHYS 381 - Electromagnetic Theory 1	PHYS 372 - Quantum Mechanics A
PHYS 399 - Research Experience in Physics	PHYS 399 - Research Experience in Physics
Year 4	
Fall Term	**Winter Term**
MATH 418 - Honors Real Variables 2	MATH 542 - Fourier Analysis
MATH 506 - Complex Variables	MATH 543 - Measure Theory 2
PHYS 458 - Special and General Relativity	PHYS 530 - Statistical Mechanics
PHYS 472 - Quantum Mechanics B	PHYS 610 - Quantum Field Theory 1
PHYS 481 - Electromagnetic Theory 2	PHYS 524 - Classical Electrodynamics
MA PH 499 - Undergraduate Research Project	MA PH 499 - Undergraduate Research Project

Course Descriptions

- **MATH 117 - Honors Calculus 1** (3 units)
  Functions, continuity, and the derivative. Applications of the derivative. Extended limits and L'Hospital's rule. Prerequisites: Mathematics 30-1 and Mathematics 31, or consent of the Department. Notes: (1) This course is designed for students with at least 80 percent in Pure Mathematics 30 or Mathematics 30-1 and Mathematics 31. (2) Credit can be obtained in at most one of MATH 100, 114, 117, 134, 144, or 154. (3) Engineering students will receive a weight of 4.0 units for this course.

- **MATH 118 - Honors Calculus 2** (3 units)
  Integration and the Fundamental Theorem. Techniques and applications of integration. Derivatives and integrals of the exponential, and trigonometric functions. Introduction to infinite series. Introduction to partial derivatives. Prerequisite: MATH 117. (Students with a 100-level calculus course different from MATH 117 may be admitted with consent of the Department.) Notes: (1) Credit can be obtained in at most one MATH 101, 115, 118, 136, 146, 156 or SCI 100. (2) Engineering students will receive a weight of 4.0 units for this course.

- **MATH 127 - Honors Linear Algebra 1** (3 units)
  Linear equations; Euclidean spaces, matrices. Complex numbers and fields. Vector spaces : basis, dimension, linear transformations. Introductions to groups and rings; permutation groups. Determinants. Eigenvalues and Eigenvectors. Prerequisite: Mathematics 30-1. Note: Credit can be obtained in at most one of Math 102, 125 or 127.

- **MATH 217 - Honors Advanced Calculus 1** (3 units)
  Axiomatic development of the real number system. Topology of Rn. Sequences, limits and continuity. Multivariable calculus: differentiation and integration, including integration in spherical and polar coordinates. The differential and the chain rule. Taylor's Formula, maxima and minima. Introduction to vector field theory. Prerequisites: One of MATH 102, 125 or 127, and either MATH 118 or MATH 216. Notes: (1) MATH 216 may be accepted as corequisite with consent of the Department. (2) Engineering students will receive a weight of 4.0 units for this course.

- **MATH 227 - Honors Linear Algebra 2** (3 units)
  Quotients and direct sums. Cayley-Hamilton. Canonical Forms (diagonal and Jordan). Real and Complex inner product spaces : orthogonality, singular value decomposition. Introduction to abstract algebra : groups, rings, and modules. Homomorphisms. Prerequisite: MATH 127. 127; Students with MATH 102 or 125 require MATH 226 as a co-requisite and consent of the Department. Note: Credit can be obtained in at most one of MATH 225 or 227.

- **MATH 317 - Honors Advanced Calculus 2** (3 units)
  Implicit function theorem. Proof of the Change of Variables Theorem. Line integrals. Theorems of Green, Gauss and Stokes in their classical form. Differential forms and Stokes' Theorem in their context. Sequences and series of functions. Uniform convergence. Prerequisite: MATH 217.

- **MATH 327 - Algebra 1** (3 units)
  Basic group theory: Groups, subgroups, normal subgroups, homomorphisms, quotient groups, coset decomposition, Example: Permutation group and general linear group; basic (commutative) ring theory: Rings, subrings, homomorphisms, ideals, quotient rings, modules over rings, submodules and quotient modules, fraction field; further group theory: Groups operating on a set, Sylow theorems. Prerequisite : One of MATH 226 or MATH 227. Note: Credit can be obtained in at most one of MATH 326 and MATH 327.

- **MATH 329 - Algebra 2** (3 units)
  Field extensions, separable and normal extensions; finite Galois theory, solvable groups and equations, constructions by ruler and compass, solutions by radicals. Time permitting, some of the following topics may be covered: Factorial rings and principal ideal domains, Noetherian rings and modules, Hilbert basis theorem. Prerequisite: MATH 327. Note: Credit can be obtained in at most one of MATH 326 and 329.

- **MATH 411 - Honors Complex Variables** (3 units)
  Complex number system. Analytic functions. Cauchy's Integral theorem and formula. Applications including the maximum modulus principle, Taylor expansion and Laurent expansion. Harmonic functions. The residue theorem with applications; calculus of residues, argument principle, and Rouche's theorem. Basics of analytic continuation. Additional topics at the instructor's discretion such as: Normal families, The Riemann mapping Theorem, Picard's Theorem. Prerequisite: MATH 314 or 317. Notes: (1) This course is primarily for Honors

students in Mathematics or Physics. (2) Offered in alternate years. It may be offered in intervening years if demand is sufficient.

- **MATH 417 - Honors Real Variables 1** (3 units)
  Brief review of set operations and countable sets. Measure theory, integration theory, Lebesgue measure and integrals on $\mathbb{R}^n$, product measure, Tonelli-Fubini theorem. Functions of bounded variation, absolutely continuous functions. Prerequisite: MATH 317 or 414.

- **MATH 418 - Honors Real Variables 2** (3 units)
  Classical Banach spaces. Hahn-Banach, open mapping and closed graphs theorems. Hilbert spaces, orthonormal bases. Elements of spectral theory, spectra of compact operators, spectral theorem for compact self-adjoint operators. Prerequisite: MATH 417. Corequisite: MATH 447 or consent of the Department.

- **MATH 432 - Intermediate Differential Equations** (3 units)
  Elementary existence and uniqueness theorems. Systems of equations, stability, perturbation theory. Introduction to numerical methods. Introduction to phase plane analysis. Prerequisite: One of MATH 201, MATH 334, MATH 336, or MA PH 251

- **MATH 447 - Elementary Topology** (3 units)
  General point-set topology. Compactness, Tychonoff's theorem, connectedness. Metric spaces, completeness, Baire's theorem. Urysohn's lemma. Topological manifolds. Homotopy theory, fundamental group, covering spaces at the discretion of the instructor. Prerequisite: MATH 216 or MATH 217. Offered in alternate years. It may be offered in intervening years if demand is sufficient.

- **MATH 448 - Introduction to Differential Geometry** (3 units)
  Topological and smooth manifolds; continuous, smooth, and invertible maps between manifolds; submanifolds; vector fields and their flows; Lie groups and algebras; left-invariant vector fields on a Lie group; differential forms, vector bundles and tensor fields; integration on manifolds; de Rham cohomology. Time permitting, additional topics at the discretion of the instructor. Prerequisite: one of MATH 209, 215, 217, 315 or MA PH 351, and one of MATH 225 or 227.

- **MATH 506 - Complex Variables** (3 units)
  A review and some extensions of single variable complex analysis. Complex linearity and holomorphicity in several variables, Hartog's theorem, Weierstrass preparation theorem, Riemann extension theorem, Weierstrass division theorem, analytic Nullstellensatz, implicit and inverse function theorems, complex manifolds and analytic subvarieties, meromorphic maps. Prerequisite: MATH 411.

- **MATH 542 - Fourier Analysis** (3 units)
  Review, theory and extension of Fourier series for square integrable functions; orthonormal systems, Bessel's inequality, completeness, Parseval's identity, Riesz-Fischer Theorem. Extension to Fourier series for functions in other Lebesgue classes; Fejer means, conjugate series, Dirichlet, Fejer and Poisson kernels. Norm convergence; remarks on pointwise convergence. Fourier transforms and series in several dimensions; inverse transform, Plancherel formula, Poisson Formula, maximal functions, Riesz-Thorin Theorem and applications. Elementary distribution theory; D, D', S, S' and some elementary results, Fourier transforms of tempered distributions. Examination of some earlier results with tempered distributions instead of functions and getting familiar with basic concepts. Prerequisite: MATH 418.

- **MATH 543 - Measure Theory 2** (3 units)
  Review of basic measure and integration theory. Signed and complex measures. Hahn and Jordan decompositions. The Radon-Nikodym theorem. Lebesgue decomposition. The Lebesgue-Stieltjes integral. Measure theory over locally compact Hausdorff spaces, in particular, the Riesz representation theorem and Haar measures. Hausdorff measure. Introduction to martingales. Prerequisite: One of MATH 417 or MATH 514, and MATH 447 or equivalent.

- **MA PH 251 - Differential Equations for Physics** (3 units)
  Differential equations occur throughout physics and being able to solve them is a critical mathematical skill for physicists. The first part of the course emphasizes solution techniques to first-order and linear, second-order ordinary differential equations, including series and Frobenius solutions, and an introduction to Fourier and orthogonal series and Sturm-Liouville problems. The second part of the course introduces partial differential equations with a study of quasilinear first-order equations, and the linear second-order wave, heat and Laplace equations, and solution techniques including the method of characteristics and separation of variables. Examples

from physics will be emphasized throughout. Prerequisite: MATH 146 or equivalent and one of MATH 102 or 125 or 127. Corequisite: MATH 214 or 217. Note: Credit may be obtained for only one of MA PH 251, MATH 201, MATH 334 or MATH 336.

- **MA PH 343 - Classical Mechanics 2** (3 units)
  Principles of mechanics; non-inertial frames; Lagrange's equations and Hamilton's principle; dynamics of oscillating systems; rigid body kinematics and dynamics; Hamiltonian methods and canonical transformations.

- **MA PH 364 - Group Theory in Physics** (3 units)
  Symmetries in physics; basic concepts of group theory and representation theory; finite groups; continuous groups; orthogonal and unitary groups; Lie groups; spinor representations; Lorentz and Poincare groups. Prerequisite: MATH 225 or MATH 227. Note: Credit can be obtained in at most one of MA PH 364 and MA PH 464.

- **MA PH 451 - Mathematical Methods for Physics 2** (3 units)
  Application to problems in physics of method of steepest descent, Fourier and Laplace transforms; boundary-value problems, integral equations, and Green's functions. Prerequisites: either MA PH 351 or both of MATH 337 and MATH 311 or MATH 411.

- **MA PH 499 - Undergraduate Research Project** (3 units)
  Undergraduate physics research project under the direction of a faculty member. Projects must involve a strong physics connection and involve some original research component. Prerequisites: A 300-level PHYS course and consent of the department. This course may be repeated but a student may obtain at most 6 units in PHYS 499 and MA PH 499.

- **PHYS 144 - Newtonian Mechanics** (3 units)
  This calculus-based course introduces the fundamentals of classical mechanics: general kinematics, Newtonian dynamics, work-energy and energy conservation, impulse-momentum and momentum conservation, rotational kinematics and dynamics, rigid bodies, angular momentum and its conservation. The course concludes with a discussion of the elastic deformation of solids and simple fluid mechanics. Prerequisites: Mathematics 30-1 and Physics 30. Mathematics 31 is strongly recommended. Corequisites: MATH 117 or 144. Note: Credit may be obtained for only one of PHYS 124, 144 or EN PH 131.

- **PHYS 181 - Relativity, Electricity and Magnetism** (3 units)
  This calculus-based course covers introductory electromagnetism including: electric forces and fields, electric potential, capacitance, DC circuits, magnetic forces and fields, magnetic fields from currents, Faraday's Law of induction, electromagnetism, and light. This leads to the breakdown of the Galilean concepts of space and time near the speed of light and an introduction to Einstein's Special Relativity including: time dilation, length contraction, Lorentz transformations, relativistic energy and momentum, Doppler effect and 4-vectors. Prerequisite: PHYS 124 (see Note following) or 144 or EN PH 131. Corequisite: One of MATH 118 or 146. Credit may be obtained for only one of PHYS 181 or 230 or 281. Note: To proceed to PHYS 181 after taking PHYS 124, it is strongly recommended that a minimum grade of B- be achieved in PHYS 124.

- **PHYS 234 - Introductory Computational Physics** (3 units)
  An introductory course on using computer based methods to solve physics problems, especially those that do not have analytical solutions or require great effort to find it. Examples of problems are drawn from mechanics, electricity and magnetism, modern physics, experimental physics, and data analysis. The course begins with an introduction to scientific programming. The topics that are covered include numerical differentiation and integration; vector geometry and linear algebra; solutions to ordinary differential equations including nonlinear equations and coupled systems of equations. Other topics will be selected from numerical methods and algorithms for analysis of physics data including root finding methods, interpolation, uncertainty estimates, an introduction to regression, Monte Carlo methods, common statistical distributions encountered in physics, Fourier analysis, signal processing and eigenvalue methods. Prerequisite: PHYS 146 or PHYS 181; MATH 118 or 146. Note: MA PH 251 or MATH 334 is a suggested corequisite.

- **PHYS 244 - Classical Mechanics 1** (3 units)
  An intermediate course in classical mechanics, which covers several applications that are critical to a wide variety of different fields of physics including: simple, damped, and driven harmonic oscillators; coupled oscillators, conservative forces, and energy. The central part of the course introduces the calculus of variations and Lagrangian mechanics, including its application to the two-body, central force problem and Noether's theorem, which connects symmetries with conservation laws. Hamiltonian mechanics is also briefly introduced. The course concludes with

the application of the linear wave equation to mechanical waves: the superposition principle, wave interference, and standing waves. Prerequisite: PHYS 124 or PHYS 144 or EN PH 131. Corequisites: MATH 102 or 125 or 127 or equivalent; MA PH 251 or MATH 201 or MATH 334 or MATH 336.

- **PHYS 271 - Introduction to Modern Physics** (3 units)
  This course covers the emergence of modern physics through revisions to the Newtonian worldview and the development of quantum mechanics. The course starts with the discovery of the wave nature of light through diffraction and interferometry leading to the observation of lines in atomic spectra. The course then discusses the early experiments that lead to the discovery of the structure of matter and early quantum phenomena including: Rutherford scattering, quantization of charge and energy, Blackbody radiation, Compton scattering, Bohr atom and de Broglie wavelength. This is followed by an introduction to the Schrödinger equation and solutions to 1D problems including: infinite and finite square potential wells, the quantum harmonic oscillator and quantum tunneling, before discussing quantized angular momentum. The course then concludes with a tour of the exciting applications of modern physics in different fields with some possible examples including: semiconductors, superconductors, nuclear decays and reactions, the Standard Model, the Higgs boson, quantum information, supernovae and Black Holes, Dark Matter, the Big Bang and Gravitational Waves. Prerequisite: one of PHYS 124, PHYS 144, or EN PH 131. Corequisites: MA PH 251 or MATH 201 or MATH 334 or MATH 336 and MATH 102 or 125 or 127.

- **PHYS 295 - Experimental Physics 1** (3 units)
  Contemporary methods of experimental physics with measurements from classical and modern physics. Analysis and graphing of experimental data using programming techniques. Estimation and statistical treatment of experimental uncertainties consistent with standard practice in physics. Planning and record keeping for experimental work, written presentation of laboratory results. Prerequisites: MATH 101 or 115 or 118 or 146, one of PHYS 124, PHYS 144, or EN PH 131; and one of PHYS 126, PHYS 146, PHYS 181, or PHYS 130. Note: To proceed to PHYS 295 after taking PHYS 126 a minimum grade of B+ in PHYS 126 and some experience of computer programming are strongly recommended. Students may only receive credit in one of PHYS 294 or PHYS 295.

- **PHYS 310 - Thermodynamics and Kinetic Theory** (3 units)
  Temperature: heat, work, and the first law of thermodynamics; entropy and the second law, enthalpy, Helmholtz and Gibbs free energy; thermodynamic equilibrium criteria; Maxwell's relations, phase transitions; elementary kinetic theory of gases. Prerequisites: one of PHYS 124, PHYS 144, or EN PH 131, and one of PHYS 126, PHYS 146, PHYS 181, or PHYS 130. Corequisite: MATH 209 or 215 or 315 or 317 or MA PH 351 or equivalent.

- **PHYS 311 - Statistical Physics** (3 units)
  Quantum states, probability distributions, temperature and entropy; canonical ensemble and the partition function; ideal gases, paramagnets; blackbody radiation. Debye model for phonons; quantum statistics; Fermi-Dirac distribution and electrons in metals; Bose-Einstein distribution. Prerequisites: PHYS 310 (or CH E 243 for Engineering Physics Program students), PHYS 271 and MATH 209 or 215 or 315 or 317 or MA PH 351 equivalent.

- **PHYS 362 - Optical Physics** (3 units)
  Electromagnetic waves; polarization and birefringence; dispersion of light in materials; Fresnel equations for reflection and transmission; multiple-beam interference; Fraunhofer and Fresnel diffraction; Fourier optics; coherence theory; optical imaging and instruments; introduction to laser physics, quantum optics, and selected current topics. Prerequisite: PHYS 181 or 230 or 281, and MATH 209 or 215 or 315 or 317 or MA PH 351.

- **PHYS 372 - Quantum Mechanics A** (3 units)
  Origins of quantum mechanics; wave functions; Schrodinger equation and its application to one dimensional systems, postulates and physical interpretation of quantum mechanics; orbital angular momentum, central potentials and three-dimensional systems. Prerequisites: PHYS 244, PHYS 271, MATH 334 or 336 or 201 or MA PH 251.

- **PHYS 381 - Electromagnetic Theory 1** (3 units)
  Review of scalar and vector fields; Gauss and Stokes theorems; curvilinear coordinates; Dirac delta function; electrostatic field and potential; electrostatic energy; conductors, capacitors; Laplace's equation; boundary value problems; methods of images; multipoles; electrostatic field in matter; polarization; displacement; linear dielectrics; magnetostatic field; Biot-Savart and Ampere's law; vector potential; magnetostatic field in matter; magnetization; linear and nonlinear magnetic media. Prerequisites: PHYS 181 or 230 or 281; MA PH 251 or MATH 201 or 337; MA PH 351 or MATH 209 or 315 or 317.

- **PHYS 399 - Research Experience in Physics** (3 units)
  A credit/no-credit course for developing research skills and participating in a research project under the direction

of a member of the Department. Prerequisites: One of PHYS 295 or PHYS 234. Two of BIOPH 201, PHYS 244, PHYS 271, PHYS 297, MA PH 251, MA PH 351. Corequisite: One of GEOPH 326, GEOPH 325, PHYS 310, PHYS 311, PHYS 362, PHYS 372, PHYS 381. Students cannot obtain credit for PHYS 399 if they have previously earned credit in or are concurrently enrolled in PHYS 499 or MA PH 499.

- **PHYS 458 - Special and General Relativity** (3 units)
  Special Relativity: space-time; Lorentz transformations; definitions of scalars, vectors and tensors; motion of a relativistic particle; energymomentum tensor and equations of motion; transformation of electromagnetic fields. General Relativity: geometry of curved space-time; equivalence principle; gravity as curvature; Einstein equations; black hole and cosmological solutions; gravitational waves. Prerequisites: MA PH 251 or MATH 337 or ECE 341, PHYS 244. Corequisite: PHYS 481.

- **PHYS 472 - Quantum Mechanics B** (3 units)
  Review of the postulates of quantum mechanics; quantization of angular momentum; matrix representations, spin and parity; approximation methods; perturbation theory; variational and other methods; applications; scattering theory; systems of identical particles. Prerequisites: PHYS 372, and MA PH 251 or MATH 337 or ECE 341 or equivalent, and MATH 311 or 411 or MA PH 351.

- **PHYS 481 - Electromagnetic Theory 2** (3 units)
  Electromotive force; Faraday's law; inductance; Maxwell's equations in free space and in matter; electromagnetic potentials; gauges; energy and momentum conservation laws; plane waves in vacuum, in nonconducting and in conducting media; reflection and refraction of electromagnetic waves; dispersion, wave guides; dipole radiation; radiation due to moving charge; radiation reaction. Prerequisite: PHYS 381, and MA PH 251 or MATH 337 or ECE 341 or equivalent.

- **PHYS 524 - Classical Electrodynamics** (3 units)
  Physical basis and foundations of Maxwell's equations, time-varying fields and conservation laws, wave equation and Green's functions, scattering and diffraction, spacetime and electromagnetism, energy-momentum of electromagnetic fields, covariant form of relativistic particle dynamics, radiation sources and damping, relativistic treatment of radiation. Knowledge of electrodynamics at the level of PHYS 481 or equivalent is required.

- **PHYS 530 - Statistical Mechanics** (3 units)
  Fundamentals of classical and quantum statistical mechanics. Density matrix methods. Micro-canonical, canonical, and grand canonical ensemble. Formulation of quantum statistics; theory of simple gases; Maxwell-Boltzmann, ideal Bose, and ideal Fermi systems. Treatment of interacting systems. Phase transitions and fluctuations. Knowledge of statistical mechanics at the level of PHYS 311 or equivalent is required.

- **BIOPH 201 - Introduction to Biophysics** (3 units)
  Physical principles important to the operation of biological systems. Biological applications of free energy, entropy, random walks, and diffusion; dynamics at low Reynolds number; cooperativity and 2-state systems; structural self-assembly; kinetic modeling; molecular motors and enzymes; membranes and potentials; genetic networks; sequences and evolution. Prerequisites: MATH 100/114/117/134/144, PHYS 124/144 or EN PH131. PHYS 126/130/146 recommended.

- **CMPUT 174 - Introduction to the Foundations of Computation 1** (3 units)
  CMPUT 174 and CMPUT 175 use a problem-driven approach to introduce the fundamental ideas of Computing Science. Emphasis is on the underlying process behind the solution, independent of programming language or style. Basic notions of state, control flow, data structures, recursion, modularization, and testing are introduced through solving simple problems in a variety of domains such as text analysis, map navigation, game search, simulation, and cryptography. Students learn to program by reading and modifying existing programs as well as writing new ones. No prior programming experience is necessary. Prerequisite: Math 30, 30-1, or 30-2. Credit cannot be obtained for both CMPUT 174 and CMPUT 274. Credit cannot be obtained for both CMPUT 174 and ENCMP 100.

- **CMPUT 175 - Introduction to the Foundations of Computation 2** (3 units)
  A continuation of CMPUT 174, revisiting topics of greater depth and complexity. More sophisticated notions such as objects, functional programming, and Abstract Data Types are explored. Various algorithms, including popular searching and sorting algorithms, are studied and compared in terms of time and space efficiency. Upon completion of this two course sequence, students from any discipline should be able to build programs to solve basic problems in their area, and will be prepared to take more advanced Computing Science courses. Prerequisite: CMPUT 174 or ENCMP 100. Credit cannot be obtained for both CMPUT 175 and CMPUT 275.

- **CMPUT 204 - Algorithms 1**
  The first of two courses on algorithm design and analysis, with emphasis on fundamentals of searching, sorting, and graph algorithms. Examples include divide and conquer, dynamic programming, greedy methods, backtracking, and local search methods, together with analysis techniques to estimate program efficiency. Prerequisites: CMPUT 175 or 275, and CMPUT 272; and one of MATH 100, 114, 117, 134, 144, or 154.

- **CMPUT 267 - Machine Learning 1**
  This course introduces the fundamental statistical, mathematical, and computational concepts in analyzing data. The goal for this introductory course is to provide a solid foundation in the mathematics of machine learning, in preparation for more advanced machine learning concepts. The course focuses on univariate models, to simplify some of the mathematics and emphasize some of the underlying concepts in machine learning, including: how should one think about data, how can data be summarized, how models can be estimated from data, what sound estimation principles look like, how generalization is achieved, and how to evaluate the performance of learned models. Prerequisites: CMPUT 174 or 274, or ENCMP 100; one of MATH 100, 114, 117, 134, 144, or 154. Corequisites: CMPUT 175 or 275; CMPUT 272; MATH 102, 125, 126, or 127; one of STAT 151, 161, 181, 235, 265, SCI 151, or MATH 181.

- **CMPUT 272 - Formal Systems and Logic in Computing Science** (3 units)
  An introduction to the tools of set theory, logic, and induction, and their use in the practice of reasoning about algorithms and programs. Basic set theory; the notion of a function; counting; propositional and predicate logic and their proof systems; inductive definitions and proofs by induction; program specification and correctness. Prerequisites: CMPUT 101, 174, 175, 274, SCI 100, or ENCMP 100.

- **ENGL 102 - Introduction to Critical Analysis** (3 units)
  Introduces methods of critical analysis through a range of literature written in English, broadly conceived, from different historical periods and cultural locations. Note: Not to be taken by students with 6 units in approved junior English.

- **ENGL 103 - Case Studies in Research** (3 units)
  This variable content course introduces methods of literary research as an in-depth process through one or more case studies. Refer to the Class Schedule and the Department of English and Film Studies website for specific topics. This course cannot be repeated for credit. Note: Not to be taken by students with 6 units in approved junior English.

- **NS 201 - Indigenous Canada** (3 units)
  For students from faculties outside the Faculty of Native Studies with an interest in acquiring a basic familiarity with Indigenous/non-Indigenous relationships. Consists of a survey of historical and contemporary relationships between Indigenous peoples and newcomers, with the aim of expanding the understandings held by many Canadians about these relationships. This course will be delivered online. Not open to students with credit in NS 200. Not designed for Native Studies majors. Sections offered at an increased rate of fee assessment; refer to the Tuition and Fees page in the University Regulations sections of the Calendar.

- **ECON 101 - Introduction to Microeconomics** (3 units)
  How markets and governments determine which products are produced and how income is distributed in the Canadian economy. Not open to students with credit in ECON 204.

- **ECON 102 - Introduction to Macroeconomics** (3 units)
  Employment, inflation, international payments, monetary policy, and fiscal policy, all in the Canadian economy. Not open to students with credit in ECON 204.

- **WRS 101 - Exploring Writing**
  This workshop course focuses on both the theory and practice of the writing process to help students experience firsthand how university writers enter into rich ongoing conversations by engaging with the words and ideas of others.

- **ECE 209 - Fundamentals of Electrical Engineering** (3 units)
  Physical concepts of passive circuit elements, Kirchhoff's laws and DC circuit equations. Energy concepts, time domain analysis of AC circuits. Impedance, complex numbers and phasor algebra. AC power concepts, resonance, three phase circuits, introduction to machines. Credit may be obtained in only one of ECE 209, E E 239, ECE 202, or E E 240, unless approved by the Department.

- **STAT 151 - Introduction to Applied Statistics 1**
  Data collection and presentation, descriptive statistics. Probability distributions, sampling distributions and the central limit theorem. Point estimation and hypothesis testing. Correlation and regression analysis. Goodness of fit and contingency table. Prerequisite: Mathematics 30-1 or 30-2. Notes: (1) Credit can be obtained in at most one of STAT 151, STAT 161, and STAT 235. (2) This course may not be taken for credit if credit has been obtained in STAT 222, STAT 266, STAT 276, KIN 109, PEDS 109, PSYCH 211, PTHER 352, SCI 151, SOC 210 or SOC 320.